3 In this question, you are provided with the following solutions.

- **FA 6** which contains two cations and one anion.
- **FA 7** which contains one cation and one anion.
- **FA 8** which contains an aqueous solution of an organic compound.

You will perform tests to identify the ions in **FA 6** and **FA 7**.
Unless otherwise stated, the volumes given below are approximate and should be estimated rather than measured.
Test and identify any gases evolved.
If there is no observable change, write **no observable change.**

(a) **(i)** Carry out the following tests. Carefully record your observations in Table 3.1.

Table 3.1

Tests		Observations
1	Add 2 cm depth of **FA 6** into a test-tube. Add aqueous sodium hydroxide until there is no further change. Filter the mixture into a clean test tube. To the filtrate, add 8 drops of dilute nitric acid, dropwise with shaking.	
2	Add 2 cm depth of **FA 6** into a test-tube. Add aqueous ammonia until there is no further change. Filter the mixture into a clean test tube.	
3	Add 2 cm depth of **FA 6** into a test-tube. Add dilute nitric acid until there is no further change.	

4	Add 2 cm depth of **FA 6** into a test-tube. Add aqueous barium nitrate until there is no further change. Add dilute nitric acid until there is no further change.	

[4]

M32 M33 M34 M35

(a) **(ii)** Carry out the following tests. Carefully record your observations in Table 3.2.

Table 3.2

Tests		Observations
5	Add 2 cm depth of **FA 7** into a test-tube. Add aqueous sodium hydroxide until there is no further change. Warm the mixture.	
6	Add 2 cm depth of **FA 7** into a test-tube. Add aqueous barium nitrate until there is no further change. Add dilute nitric acid until there is no further change.	
7	Add 2 cm depth of **FA 7** to a test-tube. Add a 1 cm depth of aqueous silver nitrate. Add aqueous ammonia to the mixture, with shaking, until the aqueous ammonia is in excess. Filter if necessary and use the resulting filtrate for **Test 8**.	
8	Add 1 cm^3 of the resulting filtrate from **Test 7** into a clean and dry boiling tube. Add 1 drop of sodium hydroxide and 10 drops of **FA 8.** Shake well and warm the mixture in a beaker of hot water for five minutes.	

[3]

M36 M37 M38

Use

3 **(b)** **(i)** Identify the ions present in **FA 6** and **FA 7**.

Cations in **FA 6**: ……. and …….

Anion in **FA 6**: …….

Cation in **FA 7**: …….

Anion in **FA 7**: ……. **[3]**

M39 M40 M41

(ii) Explain, with the aid of a suitable equation, the observations when dilute nitric acid was added to the resulting filtrate in **Test 1**.

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…… **[2]**

M42 M43

(c) **(i)** Based on your observations in **Test 8**, identify the functional group present in **FA 8**.

Functional group present in **FA 8**: …… **[1]**

M44

(ii) Briefly explain the role of the resulting filtrate from **Test 7** in **Test 8**.

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……

…… **[1]**

M45

[Total: 14]